

# **Incorporating Domain Knowledge in Multi-Fidelity Machine Learning Methods**

## **Final Report – Undergraduate Thesis**

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## Glossary

|             |                                               |
|-------------|-----------------------------------------------|
| <b>ARD</b>  | Automatic Relevance Determination             |
| <b>CMAB</b> | Contextual Multi-Armed Bandit                 |
| <b>ELBO</b> | Evidence Lower Bound                          |
| <b>GP</b>   | Gaussian Process                              |
| <b>LLM</b>  | Large Language Model                          |
| <b>MBPP</b> | Mostly Basic Python Problems                  |
| <b>MMLU</b> | Massive Multitask Language Understanding      |
| <b>NLP</b>  | Natural Language Processing                   |
| <b>PCA</b>  | Principal Component Analysis                  |
| <b>RBF</b>  | Radial Basis Function                         |
| <b>SLM</b>  | Small Language Model                          |
| <b>SVGP</b> | Sparse Variational Gaussian Process           |
| <b>UMAP</b> | Uniform Manifold Approximation and Projection |

## Structured Abstract

### Background:

The proliferation of Large Language Models (LLMs) has introduced unprecedented capabilities in natural language processing, but at a significant computational and ecological cost. While Small Language Models (SLMs) function as a lower-latency, cost-effective alternative, they struggle to define clear boundaries of competence with complex reasoning tasks, limiting their autonomous deployment in production environments.

### Methods:

We propose a routing architecture which utilizes a Sparse Variational Gaussian Process (SVGP). By extracting high-dimensional semantic embeddings from the hidden layers of a draft SLM and applying Principal Component Analysis (PCA) for dimensionality reduction, we map regions of low confidence in the model's latent space. The SVGP treats the routing decision as a Bayesian problem, quantifying mathematical uncertainty to determine when a query should be routed to the Oracle LLM.

### Results:

The system was evaluated on the Mostly Basic Python Problems (MBPP) and Massive Multitask Language Understanding (MMLU) datasets. The empirical results demonstrate that the cost-configurable SVGP outperforms traditional Contextual Multi-Armed Bandit (CMAB) baselines, achieving a superior cost-performance trade-off by avoiding the exploration tax associated with online trial-and-error methods.

### Conclusions:

This research concludes that the internal organization of the SLM's latent space provides a highly viable basis for scalable multi-fidelity routing. This architecture provides a proven method to reduce the financial and ecological footprint of generative AI without sacrificing system reliability through the accurate quantification of SLM uncertainty before autoregressive generation begins.

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# 1 Introduction

State-of-the-art Large Language Models (LLMs) perform at a high level but incur significant computational cost and latency (Pope et al., 2022). Due to their autoregressive nature, these models require massive parameter counts and vast amounts of memory to generate text token by token. Consequently, to deploy a high-fidelity LLM for every given query can be computationally exhaustive. Conversely, Small Language Models (SLMs) offer a much greater efficiency, permitting rapid inference and local deployment; however, they typically suffer from an unreliable competence boundary when faced with complex reasoning tasks. A system that can intelligently determine when to delegate a query to a less capable, but less computationally expensive model would significantly reduce the monetary cost and carbon footprint of deploying these models, while maintaining accuracy (Chen et al., 2023; Strubell et al., 2019).

To address this computational bottleneck, many multi-fidelity routing systems have been proposed. Approaches such as Multi-Armed Bandits (MAB) (Sutton and Barto, 2018) and Speculative Decoding (Leviathan et al., 2023) help mitigate these issues, but suffer from their own operational trade-offs, such as large parallel memory overhead or a reliance on costly trial-and-error strategies for query inputs. A robust and effective multi-fidelity system should possess a gatekeeper capable of proactively detecting failure before autoregressive generation begins, minimizing both computational waste and latency.

Building upon this need for a proactive gatekeeper, we propose a multi-fidelity router that intercepts the SLM’s internal hidden state immediately after a user’s query is processed. Because modern language models process language as high-dimensional continuous vectors, we can extract this latent representation to create a dense semantic understanding of the query and the associated uncertainty of the SLM. Rather than treat the routing operation as a simple classification problem, we formulate it as a decision-making problem under uncertainty (Berger, 1985), and utilize a Gaussian Process (GP) (Rasmussen and Williams, 2006) to model the gap in utility between high and low fidelity models based purely on the derived semantic embeddings. Furthermore, since standard GPs scale poorly with high-dimensional continuous data, we incorporate Principal Component Analysis (PCA) to compress the SLM’s embeddings, isolating the core semantic meaning from the noise without a significant computational overhead.

To evaluate the performance of this architecture, two key research questions are addressed. First, can the significantly compressed, continuous latent space of an SLM retain enough meaning to accurately predict autoregressive failure prior to generation? Second, how do the trade-off curves generated by the Sparse Variational Gaussian Process-based routing system compare against the established Speculative Decoding and Contextual Multi-Armed Bandit (CMAB) baselines?

For a more thorough analysis of the proposed routing system’s competence and to reinforce the correctness of the methodology, we adopt a phased testing approach. Initial validation is conducted with a code-generation sandbox, utilizing a dataset of Mostly Basic Python Problems (MBPP) (Austin et al., 2021). The binary result of executing code allows for an objective success metric for our training data, permitting a more rigorous verification of the router’s predictive accuracy. Following this validation phase, the system is generalized across broader natural language processing tasks. For this form of testing, we utilize the Massive Multitask Language Understanding (MMLU) benchmark (Hendrycks et al., 2020), which contains multiple-choice questions across 57 academic and professional subjects. This allows for the router to be evaluated on a spectrum of semantic domains and task difficulty levels.

The empirical results of this research demonstrate that the SVGP router successfully navigates the multi-fidelity trade-off. In both the objective code generation and generalized Natural Language Processing (NLP) task benchmarks, the SVGP router performed at near-Oracle accuracy while providing a reduction in aggregate computational cost. The novelty and significance of this architecture lie in its proactive methodology: existing baselines rely on rigid parallel execution or post-generation trial-and-error; however, this work proves that model uncertainty can be accurately quantified from purely the geometric distribution of the semantic embeddings prior to any autoregressive generation.

The remainder of the thesis is structured as follows: Section 2 provides an explanation of foundational background concepts and related work, outlining the current research gap in multi-fidelity systems. Section 3 describes the primary research goal of the project and the specific objectives associated with that goal. Section 4 details the methodology, including the router’s architecture and the extraction of latent states. Section 5 presents and analyzes the empirical results and performance trade-off curves across the MBPP and MMLU datasets. Section 6 discusses what implications, limitations, and threats to the validity of the findings exist. Section 7 summarizes the conclusions drawn from the research. Lastly, Section 8 highlights key insights gained from the research and what future work may follow.

## 2 Background and Related Work

This section introduces the machine learning concepts and foundational mathematics required for the operation and success of the proposed multi-fidelity routing system. First, we review probabilistic reasoning and neural network optimization. Then, we detail the inner workings of Large Language Models, Multi-Armed Bandits, and Gaussian Processes.

### 2.1 Large Language Models

Large Language Models are deep neural networks built on the Transformer architecture. Unlike older text models, which would read left to right, Transformers make use of a *self-attention* mechanism (Vaswani et al., 2017). This allows the model to consider the entire sequence of text and determine which words are most important to the current context. These models are trained using an objective called *next-token prediction*. The model, upon being given a sequence of words, must predict which word comes next. To increase the efficiency of this process, the model makes use of a *Key-Value (KV) Cache* (Kwon et al., 2023). Since the model generates one word at a time, it would have to re-read the entire sentence when processing each new word. However, the KV Cache resolves this problem by saving the Keys and Values (mathematical representation) of each past word in memory.

### 2.2 Speculative Decoding

Speculative Decoding is an optimization technique designed to lessen the autoregressive latency of Large Language Models. This approach employs a dual-model architecture in which a highly efficient *draft model* (an SLM) works in combination with a more capable *target model* (a high-fidelity Oracle) (Leviathan et al., 2023). In this framework, the draft model generates a sequence of

speculative tokens, of which the quantity is determined by the system designer. The target model will then process all speculative tokens in parallel during a single forward pass. For each token, the target model will compute a probability distribution. Then, a verification algorithm, typically using a form of rejection sampling, compares the computed probabilities of both the target model and the draft model. If a token from the draft model satisfies the defined acceptance criteria, it is retained; however, if a token is rejected, the target model immediately overwrites the rejected token and discards all remaining speculative tokens. Because the target model verifies each token, this system guarantees an output identical to that of the target model running on its own, while reducing overall execution latency.

### 2.3 Multi-Armed Bandit Problem

In a setting of sequential decision-making, an approach such as batch learning may degrade in performance if the incoming data stream deviates from the training distribution. Therefore, it is useful to employ options that utilize Bayesian updating to further refine the accuracy of the model as new data comes in, keeping it constantly attuned to input alterations. This scenario can be formally modeled as the MAB problem (Sutton and Barto, 2018). The three components are the *agent* (the system making the choice), the *arms* (the  $K$  options available), the *reward* (the binary outcome of success or failure). The Agent is presented with two separate approaches at all times: *exploitation* (using current knowledge to maximize short-term yield) and *exploration* (sacrificing the current yield to reduce uncertainty in other arms).

A possible solution for the difficulty of balancing these two approaches is *Thompson Sampling*. This method involves sampling a value from the posterior distribution (Beta distributions) of each arm. The arm that generates the highest sampled value is selected for execution.

Upon observing the actual reward, the selected arm's posterior distribution is updated accordingly:  $\alpha$  is incremented for a success, and  $\beta$  is incremented for a failure. This probabilistic method allows for uncertain arms to occasionally be selected due to their wide and flat posterior distributions, while strong arms are exploited more frequently due to their narrow and elevated distributions.

### 2.4 Gaussian Processes

A Gaussian Process (GP) is a collection of random variables, any finite number of which have a joint Gaussian distribution. It provides a probabilistic approach to modeling nonlinear functions while also providing the associated uncertainty. A GP is specified by its *mean function*  $m(x)$  and its *covariance function*, or *kernel*,  $k(x, x')$ :  $f(x) \sim \mathcal{GP}(m(x), k(x, x'))$ . A kernel function  $k(x, x')$  determines the “smoothness” of the functions sampled from the GP. It defines the similarity between two data points: if  $x$  and  $x'$  are close in input space, their outputs  $f(x)$  and  $f(x')$  should be highly correlated. A common choice is the *Radial Basis Function (RBF)* kernel:

$$k(x, x') = \sigma^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right).$$

The hyperparameter  $\ell$ , known as a *length scale*, is critical for incorporating domain knowledge. When the length scale is small, the distance between points is mathematically amplified, meaning

the covariance, and therefore the correlation, between  $f(x)$  and  $f(x')$  decreases at a very rapid rate as they move apart. As a result of this, the GP expects a function to change rapidly, creating highly volatile, jagged decision boundaries. Conversely, a large length scale implies the data points will remain strongly correlated even when separated by a great distance, yielding much smoother boundaries. We can leverage this hyperparameter to encode our domain knowledge regarding how quickly a model’s competence may shift as a result of changes in semantic meaning. While exact GPs offer robust uncertainty quantifications on little data, their  $O(N^3)$  computational time complexity is problematic for real-time routing, especially on large datasets. To resolve this, we utilize a *Sparse Variational Gaussian Process (SVGP)* (Titsias, 2009), which reduces the computational time complexity to  $O(NM^2)$  by employing representational inducing points to approximate the posterior distribution. For a more comprehensive mathematical formulation of the SVGP, refer to Appendix A.

## 2.5 Analysis and Research Gap

Existing multi-fidelity routing systems are able to address the computational bottlenecks of large language models; however, they introduce significant operational trade-offs. Speculative decoding (Leviathan et al., 2023) preserves high accuracy while reducing latency, but it incurs a heavy memory and compute overhead by requiring parallel operation of both the draft and target models. Conversely, Contextual Multi-Armed Bandit (CMAB) (Sutton and Barto, 2018) architectures avoid parallel execution cost but treat the draft model as a black box. Due to CMAB’s reliance on online trial-and-error for mapping competence boundaries, it suffers from a persistent “exploration tax”, forcing the system to degrade aggregate accuracy by learning through failure.

The current literature lacks a proactive, transparent routing system capable of predicting a Small Language Model’s failure with high accuracy before the autoregressive generation begins. We address this crucial research gap by extracting a feature vector from the continuous latent space of the draft model immediately after a prompt is processed, and applying PCA (Pearson, 1901) to compress these high-dimensional embeddings. We then employ a SVGP (Titsias, 2009) to map the decision boundaries directly from these embeddings, proactively evaluating model confidence. This routing architecture achieves high accuracy and computational efficiency, while effectively circumventing both the rigid memory overhead of Speculative Decoding as well as the inherent exploration tax of CMAB systems.

## 3 Research Goal and Objectives

The primary goal of this thesis is to develop a routing system that dynamically selects the most cost-effective machine learning model for a given input, while maintaining a high level of accuracy. The system utilizes a Gaussian Process to quantify the uncertainty and risk of the routing decisions. To attain this goal, our research is structured around four specific objectives that progress from theoretical mathematical formalization to practical system evaluations:

- **O1:** The formalization of the multi-fidelity routing problem as a Gaussian Process Regression task. The semantic features of the latent space are mapped using domain knowledge encoded directly into the prior distributions.

- **O2:** The implementation of a SVGP to map the decision boundaries within the latent space of a Small Language Model.
- **O3:** A comparison demonstrating the trade-off curves (computational cost and latency vs. accuracy) for the SVGP router against the CMAB and Speculative Decoding baselines, utilizing the MBPP and MMLU datasets.
- **O4:** An evaluation framework utilizing pointwise binary classification via an Oracle LLM judge, which minimizes the subjective scoring bias by evaluating multiple choice answers against ground truth labels.

## 4 Methodology

This section details the implementation of our proposed multi-fidelity routing system. In a transition from theoretical concepts of probabilistic modeling, we will outline, from end-to-end, the architecture designed to intercept, analyze, and route queries based on the continuous latent space of a Small Language Model.

### 4.1 Data Acquisition and Feature Extraction

First, we validated the system using the MBPP (Austin et al., 2021) dataset. This provided an objective baseline, as the outputs generated by the draft model (the SLM) could be executed within a Python sandbox. If the generated code passed the execution test, it was labeled a success (1); otherwise, it was labeled a failure (0). Once this testing stage was completed, the system’s capabilities were generalized using the MMLU (Hendrycks et al., 2020) dataset. For these queries, binary labels were generated through a form of pointwise evaluation, in which an Oracle judge strictly compared the selected answer of the SLM against the ground-truth label, assigning a deterministic success or failure outcome label.

An additional challenge is introduced when evaluating the absolute capability of these multi-fidelity systems on NLP queries. The standard evaluation protocol will often rely on an LLM-as-a-judge format, in which a numerical score (e.g., grading an output from 1 to 10) is assigned to each individual execution. This methodology, however, can suffer from a significant computational overhead, high variance, and a calibration bias (LLMs are generally reluctant to give extreme scores such as, in this context, 1 and 10). To ensure a more stable metric and reduce evaluation costs, this thesis avoids open-ended grading entirely by utilizing the multiple-choice format of the MMLU dataset. An Oracle LLM strictly evaluates the text generated from the draft model against the ground-truth choice (A,B,C, or D), assigning a 1 (Success) or 0 (Failure). This binary evaluation framework satisfies **O4**, utilizing ground truth label comparison to minimize subjective scoring bias. With the dataset of prompts established, the critical step was extracting the continuous latent representations from the SLM to utilize as input features  $X$  for the SVGP. In modern autoregressive language models, the input prompts are processed by mapping the discrete tokens to a continuous vector space.

We intercept this representation utilizing the Hugging Face Transformers library, configuring the SLM to output its internal hidden state values during its forward pass and halting prior to token generation. Specifically, the final Transformer layer is our target. Due to the model’s causal

---

```

1  # Turning off gradient tracking to preserve memory
2  with torch.no_grad():
3      # Performing a forward pass on the input
4      outputs = model(**inputs)
5
6      # Extracting the hidden state from the final token of the prompt
7      # Moving to CPU and casting to float32 for PCA compatibility
8      hidden_state = outputs.hidden_states[-1][0, -1, :] \
9          .cpu().numpy().astype(np.float32)

```

---

**Listing 1:** Extracting the final continuous hidden state from the SLM.

self-attention mechanism, the hidden state of the final token in the prompt sequence contains the semantic context of the preceding tokens in the query.

The high-dimensional vector, representing the complete semantic embedding of the user's query, is extracted and paired with the respective binary outcome label ( $y \in \{0, 1\}$ ). This forms a collection of  $(X, y)$  pairs, which are foundational training data required to map the SLM competence boundaries within the latent space.

## 4.2 Dimensionality Reduction

Upon completion of the data acquisition phase, each query is represented by a 960-dimensional continuous vector. While the density of this embedding successfully captures the rich semantic meaning of the query, it also acts as a significant mathematical bottleneck for probabilistic modeling.

Gaussian Processes are highly susceptible to a phenomenon called the *curse of dimensionality*. Due to the RBF kernel computing the Euclidean distance to determine the covariance between points, operating in a 960-dimensional space dilutes the concept of proximity. In such a high-dimensional space, the distance between any two random points trends towards uniformity, making it very difficult for the GP to accurately draw distinct boundaries of competence for the SLM. Further, the computational complexity of a standard kernel operation scales poorly as the feature space grows.

The raw embeddings  $X_{\text{raw}}$  must be compressed into a lower-dimensional latent space before the routing occurs in order to resolve this issue. This can be done utilizing *Principal Component Analysis* to represent the 960-dimensional vectors as highly compact 10-dimensional representations  $X_{\text{pca}}$ .

PCA calculates the covariance matrix of the whole dataset and isolates the primary axes that capture the highest degree of variance within the data. The system preserves the most critical information by projecting the original embeddings onto the top 10 axes, which contain the most significant mathematical spread. This filters out the statistical and linguistic noise inherent in language models while also retaining the core semantic shifts needed for the Sparse Variational Gaussian Process to successfully classify queries.

### 4.3 Model Training and Technologies

Once the dimensional reduction is complete, the 10-dimensional embeddings  $X_{pca}$  and their associated binary labels are used to train the router. The development process utilized the PyTorch and GPyTorch libraries for hardware-accelerated tensor calculations and probabilistic modeling to satisfy the computational requirements of the pipeline. The model was implemented as a SVGP. The system was configured with  $M = 100$  dynamically configurable inducing points; this ensures efficient real-time routing. The kernel, length scales, and global variance are all optimized during the machine learning process by maximizing the *Evidence Lower Bound (ELBO)* via an Adam optimizer. For a comprehensive mathematical formulation of this training architecture, including Automatic Relevance Determination (ARD) kernel configuration and the Cholesky variational strategy, refer to Appendix A. This formalization of the multi-fidelity routing problem as a sequential decision-making task directly addresses **O1** and **O2**.

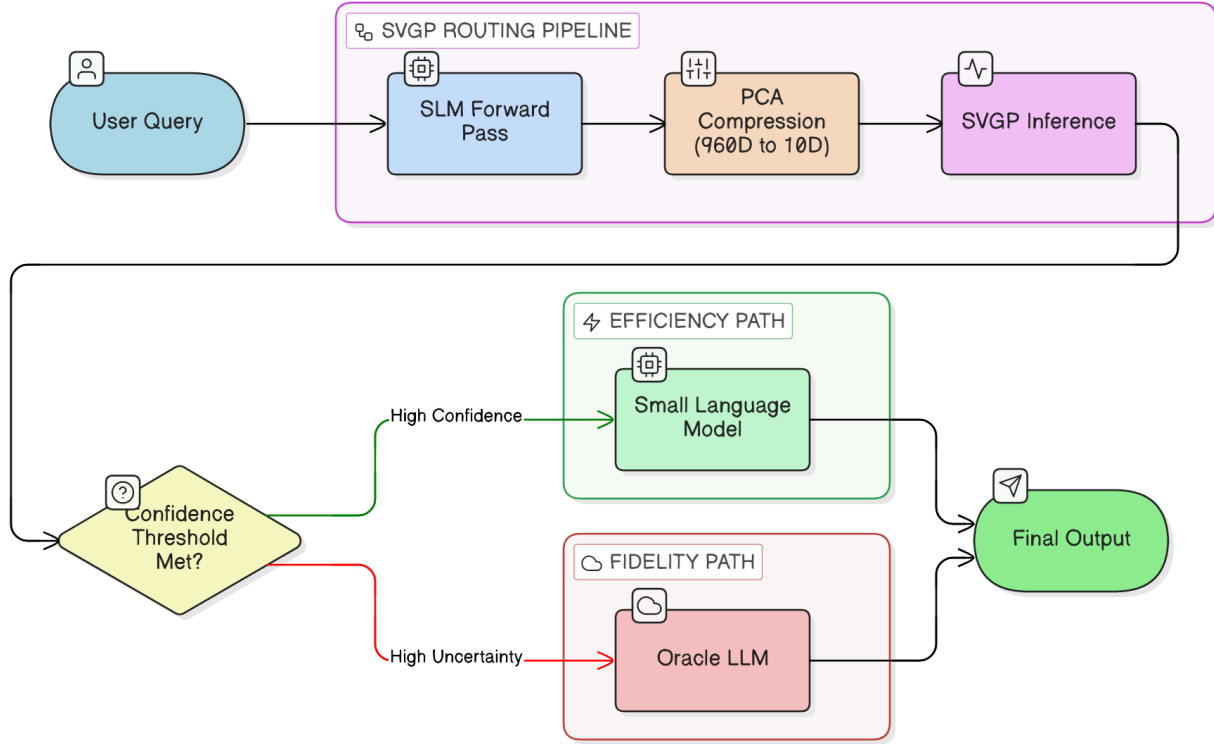
## 5 Results

To evaluate the effectiveness of the SVGP router, we evaluate how it performs against two established multi-fidelity baselines: a CMAB utilizing Thompson Sampling and Speculative Decoding. This will be done by quantifying the trade-off between overall system accuracy and aggregate computational cost. Computational cost is treated as a unified metric composed of both latency (inference time) and the computational overhead.

### 5.1 System Architecture and Design

This routing pipeline contains a sequential decision-making process. It intervenes before the autoregressive generation begins. The lifecycle of a query in the proposed system involves five distinct stages:

1. **Query Acquisition:** A user’s query is submitted to the system and processed by the draft model (the SLM)
2. **Latent State Extraction:** After the SLM completes a single forward pass, but before any tokens are generated, the final hidden state is extracted. This hidden state is a continuous vector that acts as a high-dimensional representation of the semantic embedding of the query.
3. **Dimensionality Reduction:** To prevent scaling problems in the Gaussian Process due to the high-dimensionality of the data, the extracted vector is compressed using PCA. This allows for some of the statistical noise to be discarded.
4. **Probabilistic Evaluation:** The compressed semantic vector is fed into the SVGP router. Utilizing the RBF kernel, the SVGP compares the vector against the decision boundaries it had previously mapped to output a predicted success probability, as well as a quantified measure of uncertainty.
5. **The Routing Decision:** The final execution path will be dictated by a predefined confidence threshold. If the SVGP predicts success with a high enough level of confidence, the SLM



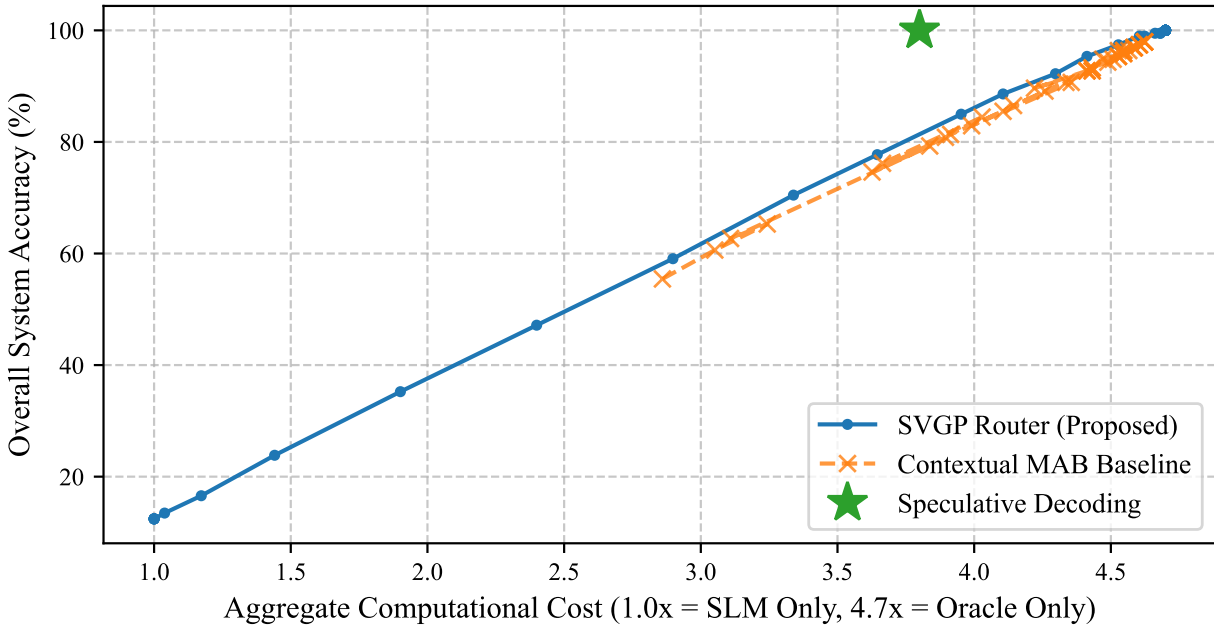
**Figure 1:** Contextual architecture of the multi-fidelity routing system, illustrating the logic path from latent state extraction to model escalation.

is authorized to complete its autoregressive generation. If the prediction provides a high probability of failure or extreme uncertainty, the generation is halted, and the query is sent to the high-fidelity Oracle LLM for execution.

## 5.2 System Implementation

With the extraction and compression of the latent representations complete, the 10-dimensional embeddings are usable for probabilistic classification training. The core component of the routing gatekeeper is a SVG P. This architecture was constructed and optimized utilizing both PyTorch and GPyTorch libraries, allowing for dynamic, hardware-accelerated mapping of the competence boundaries for the Small Language Model. We selected the RBF kernel because its reliance on Euclidean distance allows the SVG P to identify the semantic clusters of competence in our PCA-compressed 10-dimensional embeddings. Additionally, a Bernoulli likelihood was used to flatten these varying outputs into a binary routing probability without incorporating inference latency.

We verified the implementation through unit testing of the PCA and SVG P modules, followed by integration testing for the whole pipeline. To guarantee stability across updates to the model, regression testing was also conducted using the MBPP dataset to confirm that new data points did not degrade established accuracy baselines.



**Figure 2:** Performance trade-off curve on the MBPP dataset, illustrating the SVGP router’s overall system accuracy against aggregate computational cost compared to the CMAB and Speculative Decoding baselines.

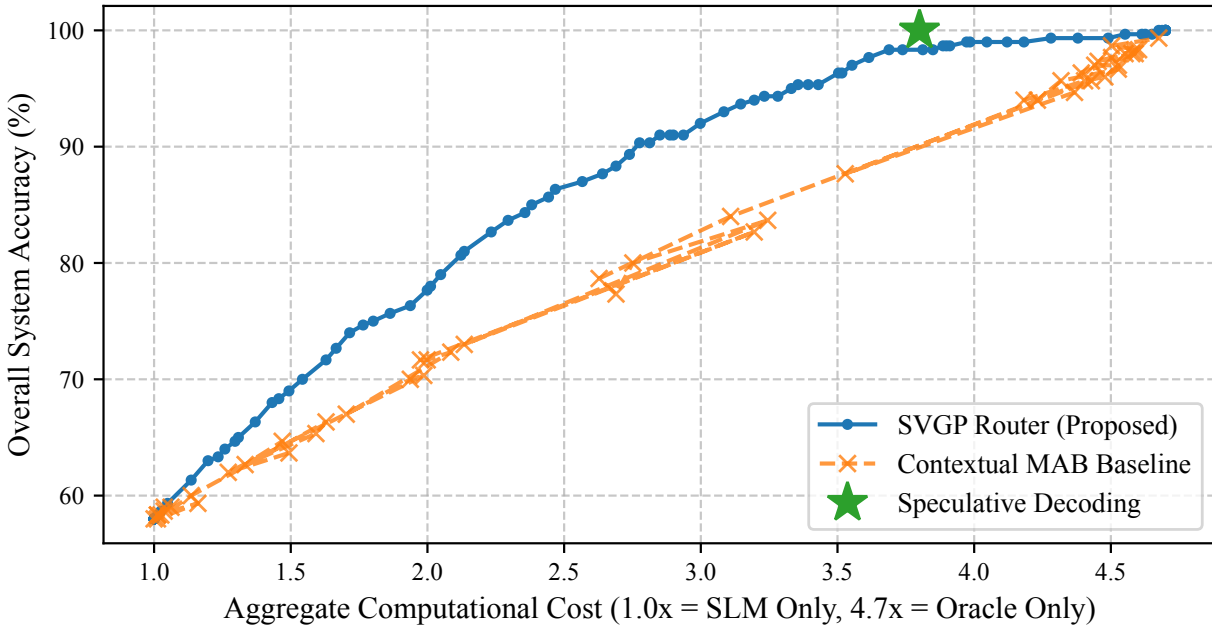
### 5.3 Objective Validation: Code Generation

The first stage of evaluation utilizes the MBPP dataset. Because this research focuses on algorithmic routing and backend compute efficiency, validation was completed via quantitative benchmarking rather than qualitative feedback from end-users or stakeholders. This dataset permits objective, binary testing: the code generated either executes successfully in an isolated Python sandbox and passes the evaluation, or it fails. By removing the subjective element of general NLP inference evaluation, this phase provides a rigorous baseline for the SVGP router’s predictive accuracy.

For this evaluation, our draft model (SLM) acts as the baseline for maximum efficiency and lowest fidelity, while the Oracle LLM acts as the baseline for maximum fidelity but highest aggregate cost. To generate these performance curves, the SVGP router’s confidence threshold was adjusted to different positions. This illustrates the balance between overall processing burden and the resulting accuracy of the system.

As Figure 2 illustrates, the SVGP router maintains a consistent efficiency advantage over the CMAB baseline across the overlapping aggregate computational cost range. Notably, the CMAB suffers from an “exploration tax” caused by its reliance on online trial-and-error for the input queries. This prevents the CMAB from operating below the aggregate computational cost threshold of  $\sim 2.8\times$ . Conversely, the proactive approach of the SVGP allows it to map the entire operational cost range, providing a routing path from the pure SLM baseline ( $1.0\times$ ) all the way to the Oracle ( $4.7\times$ ).

While the SLM’s baseline accuracy on the MBPP dataset is exceptionally low ( $\sim 12\%$ ), the SVGP permits a linear path of escalation. By querying the Oracle only when uncertainty is high, it yields a higher accuracy than the CMAB for any computational budget. Finally, although



**Figure 3:** Performance trade-off curve on the MMLU dataset, demonstrating the SVGP router’s accuracy compared to the CMAB and Speculative Decoding baselines.

Speculative Decoding provides perfect accuracy, it is completely computationally rigid, at a 3.8x cost overhead. While the SVGP performs at a lower accuracy ( $\sim 82\%$ ) at this specific point on the curve, it offers cost flexibility. For environments with strict budgets for memory or latency (e.g., the 1.5x to 3.0x cost ranges), Speculative Decoding is not an implementable approach. The SVGP thus provides the critical advantage of a dynamic threshold that can be dialed down while still providing the best available accuracy for that specific budget. These results address **O3**, demonstrating the SVGP’s trade-off curve against the CMAB and Speculative Decoding baselines.

## 5.4 Generalized Evaluation: Natural Language Processing

Upon completion of the objective validation in the code-generation environment, the SVGP router was evaluated on its ability to generalize to broader knowledge domains using the MMLU dataset. MMLU evaluates the model across 57 distinct subjects, offering a contrast to the highly structured syntax of Python code.

This dataset provides the ideal environment to benchmark the router’s sensitivity to semantic clustering. In the latent space, the difference between distinct subjects forms regional clusters of competence based on the SLM’s knowledge limits.

As demonstrated in Figure 3, the SVGP router is able to identify these semantic boundaries, yielding a convex trade-off curve which significantly outperforms that of the CMAB baseline. Due to query inputs being concentrated on subject matter, the extracted hidden states provide a strong representation of the question’s semantic domain. The Gaussian Process effectively identifies which clusters the SLM is capable of answering correctly, and which domains represent a high probability of failure.

By selectively routing the most difficult or out-of-distribution subject queries to the LLM Oracle, the system outlines an optimal path that scales towards 100% accuracy while keeping the aggregate computational cost lower than a pure deployment of the LLM Oracle. Conversely, the CMAB baseline exhibits a high variance and instability due to its reliance on online trial-and-error across a variety of subject contexts. This generalized evaluation completes the benchmark comparison of **O3** across a broad semantic domain.

## 5.5 Novelty of Results

The primary novelty in our proposed router architecture lies in how the multi-fidelity routing decision is formulated. Rather than being initialized with uninformative priors and suffering an “exploration tax” to map competence distributions as the CMAB does, the SVGP utilizes highly informative priors derived directly from the draft model’s latent space. The SVGP exploits the draft model’s pre-existing semantic awareness to proactively map competence boundaries prior to autoregressive generation. Furthermore, although lossless verification methods such as Speculative Decoding guarantee high accuracy, they introduce a rigid computational overhead. The SVGP introduces a novel mechanism for cost-flexibility: by quantifying Bayesian uncertainty directly from the latent embeddings, the system can dynamically adjust routing thresholds across the entire computational range without incurring an exploration penalty or relying on arbitrary confidence scores.

## 6 Discussion

This section interprets the broader meaning of the results and the main limitations of the proposed routing architecture. It also discusses factors that may affect the validity of the findings and how well they generalize. Together, these points provide clarity on the strengths of the approach as well as the boundaries of the conclusions.

### 6.1 Threats to Validity

The primary threat to the external validity of this research lies in the structural constraints of the evaluation datasets. Both the MMLU and MBPP benchmarks provide highly objective, measurable environments for mapping the competence boundaries of the SLM. However, they do not fully encapsulate the complexity of real-world deployments. MMLU relies on an inflexible multiple-choice format, which contrasts with the real open-ended, conversational, and sometimes ambiguous queries typical of a human-computer interaction. Similarly, although MBPP provides foundational programming tasks, it is not reflective of the multi-file, repository-level coding environment required in professional software development. Consequently, the SVGP’s impressive performance on these controlled benchmarks may not perfectly translate to real-world implementations.

A secondary threat concerns the validity of the chosen evaluation framework for the MMLU benchmark, and how it relies on an Oracle LLM as the judge for ground-truth labels. Although the methodology limits subjective scoring bias by restricting the Oracle to binary evaluations of multiple-choice answers (A, B, C, or D), utilizing an LLM as a judge introduces the risk of algorithmic bias. The model’s pre-training may cause it to favor certain text patterns or structural

formats generated by the SLM, skewing the success and failure labels. To mitigate subjective evaluation bias, the MBPP dataset was used alongside the MMLU dataset, providing a binary pass/fail execution metric that is not influenced by LLM interpretation. Nevertheless, the latent preferences of the Oracle model remain a residual threat to the absolute accuracy of the classifications for the MMLU dataset.

## 6.2 Implications of the Research

The findings of this research carry significant implications for the economic and environmental influence of generative AI. Widespread integration of LLMs continues to scale, and the financial overhead, paired with the massive energy consumption associated with autoregressive inference, is becoming unsustainable for many organizations. This Sparse Variational Gaussian Process routing approach can proactively filter and route queries with high accuracy, providing a viable pathway to largely reduce reliance on high-fidelity cloud models. Unnecessary compute waste is minimized by routing foundational queries to highly efficient draft models, directly addressing the ecological and financial bottlenecks imposed on modern NLP systems.

Further, these results have implications for localized AI deployment. SLMs are highly desirable for deployment on consumer-grade hardware due to their privacy benefits and low latency. However, unpredictable failure rates have limited their autonomous use. The SVGP router provides a reliable, lightweight gatekeeping framework for these local models, allowing safe deployment. The system ensures that user queries are only processed locally when the model is confident by mathematically quantifying uncertainty prior to execution, allowing for a seamless escalation to a high-fidelity Oracle only when required.

## 6.3 Limitations

The SVGP’s primary limitation is its dependency on the representational quality of the draft model’s latent space. If the SLM lacks a sufficient amount of pre-training on a domain, the extracted hidden states will likely not contain enough semantic signal for the Gaussian Process to draw the competence boundaries accurately. Furthermore, though PCA effectively reduces dimensionality, it may fail to capture highly non-linear relationships within the embeddings that may be identifiable to a more complex (but computationally expensive) encoder. These limitations can be minimized in future iterations by substituting PCA with non-linear dimensionality reduction techniques such as Uniform Manifold Approximation and Projection (UMAP).

## 6.4 Generalizability

Our SVGP multi-fidelity routing architecture demonstrates a high degree of generalisability across different domains and models. Because the system relies primarily on continuous hidden states, a fundamental feature of all Transformer-based language models, the SVGP is not constrained to a specific SLM. It can be adapted to other open-weight models with only a retraining of the Gaussian Process on the new embeddings, requiring minimal architectural overhead. Further to this, although the underlying probabilistic framework is only validated on code generation and academic multiple-choice benchmarks, it is theoretically applicable to any autoregressive task where distinct competence boundaries can be mapped within the semantic embeddings.

## 7 Conclusions

Unprecedented capabilities in natural language processing have been unlocked by the proliferation of Large Language Models. However, this has incurred a severe computational and ecological cost (Strubell et al. (2019); Chen et al. (2023)). Small Language Models are a highly efficient alternative, but their boundaries of competence are often unreliable with complex reasoning tasks. To bridge this gap, this thesis proposed a proactive, multi-fidelity routing architecture. This architecture minimizes aggregate computational overhead while ensuring high system accuracy.

By isolating the high-dimensional latent space of the draft model and compressing it via Principal Component Analysis, we successfully derive the semantic features critical to predicting a model failure. We treated the routing decision not as a classification task, but rather as a sequential decision-making problem under uncertainty. This Bayesian approach, utilizing a SVGP, mapped the draft model’s competence boundaries and quantified its mathematical uncertainty prior to autoregressive generation.

As detailed in Section 5, our empirical results, validated using the MBPP dataset (Figure 2) and the MMLU dataset (Figure 3), demonstrate the profound effectiveness of this approach. The SVGP router generated a flexible, multi-fidelity trade-off curve that outperformed the CMAB baseline in strict syntax generation and broad semantic reasoning. By leveraging a proactive analysis of the latent features, the SVGP bypassed the “exploration tax” which is inherent in online trial-and-error methods. Furthermore, the SVGP, due to its dynamic confidence threshold, proved to be a computationally adjustable alternative to Speculative Decoding, which provides lossless accuracy, but at a high, static compute cost. All four of the research objectives were satisfied: the routing problem was formalized as a Gaussian Process Regression task (**O1**), the SVGP was implemented and mapped the latent competency boundaries (**O2**), trade-off curves were produced and compared the SVGP against both baselines across MBPP and MMLU datasets (**O3**), and a deterministic Oracle framework was employed to eliminate subjective scoring bias (**O4**). Ultimately, this research concludes that the inherent geometry of a draft model’s latent space can be leveraged as a highly viable Bayesian prior, allowing for scalable multi-fidelity routing. By accurately quantifying the SLM’s uncertainty prior to autoregressive generation, this architecture provides a proven method to drastically reduce the financial and ecological strain of generative AI without sacrificing system reliability.

## 8 Future Work and Lessons Learned

This thesis establishes a robust foundation for multi-fidelity routing; however, there remains much to be researched regarding the evaluation parameters. The SVGP router was validated using a specific draft model and an objective code-generation dataset. Therefore, its performance across a wider variety of language models and problem domains is yet to be evaluated. Additionally, future work should investigate the impact of alternative dimensionality reduction techniques. While our current architecture linearly compresses the latent space to ten dimensions using PCA, implementing non-linear algorithms such as UMAP, as well as modifying the target dimension count, could better preserve the complex semantic meaning during the compression. This would provide a deep insight into optimizing Gaussian Processes and further maximizing the predictive accuracy of the router.

Critical insights were yielded from the development and evaluation of our SVGP router, in-

sights which are applicable to the broader field of natural language processing: the compressed, continuous latent space of an SLM contains sufficient structural meaning to proactively predict autoregressive failure. Traditionally, the analysis of output token probabilities (logits) or a type of post-generation verification is utilized to predict model failure. However, this research demonstrates that the model's uncertainty can be accurately quantified solely from the geometric distribution of the semantic embeddings prior to generation. By demonstrating that a 10-dimensional PCA compression preserves these crucial competence boundaries, this work proves that a proactive, low-cost router is mathematically viable without requiring access to a model's underlying parameter weights.

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## References

- Austin, J., Odena, A., Nye, M., Bosma, M., Michalewski, H., Dohan, D., Jiang, E., Cai, C., Terry, M., Le, Q., et al. (2021). Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*.
- Berger, J. O. (1985). *Statistical decision theory and Bayesian analysis*. Springer.
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
- Chen, L., Zaharia, M., and Zou, J. (2023). Frugalgpt: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., et al. (2021). Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*.
- Do, B. and Zhang, R. (2023). Multi-fidelity bayesian optimization in engineering design. *arXiv preprint arXiv:2311.13050*.
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. (2020). Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*.
- Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., and Gonzalez, J. E. (2023). Efficient memory management for large language model serving with pagedattention. In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626.
- Leviathan, Y., Kalman, M., and Matias, Y. (2023). Fast inference from transformers via speculative decoding. In *International Conference on Machine Learning*. PMLR.
- Pearson, K. (1901). Liii. on lines and planes of closest fit to systems of points in space. *The London, Edinburgh, and Dublin philosophical magazine and journal of science*, 2(11):559–572.
- Pope, R., Douglas, S., Chowdhery, A., Devlin, J., Bradbury, J., Heek, J., Xiao, K., Agrawal, S., and Dean, J. (2022). Efficiently scaling transformer inference. *arXiv preprint arXiv:2211.05102*.
- Rasmussen, C. E. and Williams, C. K. I. (2006). *Gaussian processes for machine learning*. MIT press.
- Strubell, E., Ganesh, A., and McCallum, A. (2019). Energy and policy considerations for deep learning in nlp. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3645–3650.
- Sutton, R. S. and Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press.
- Titsias, M. (2009). Variational learning of inducing variables in sparse gaussian processes. In *Artificial intelligence and statistics*, pages 567–574. PMLR.

- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in neural information processing systems*, pages 5998–6008.
- Wang, Z., Xing, W., Kirby, R., and Zhe, S. (2021). Multi-fidelity high-order gaussian processes for physical simulation. In *International Conference on Artificial Intelligence and Statistics*, pages 865–873. PMLR.

## Appendix A Mathematical Formulation of the SVGP Router

### A.1 Variational Strategy and Inducing Points

A standard exact Gaussian Process makes a prediction based on an evaluation of the covariance between all known training examples. Mathematically, this process requires an inversion of an  $N \times N$  covariance matrix, which has a computational complexity represented as  $O(N^3)$ . As the scale of the dataset grows, this exact mathematical approach quickly overwhelms available computational resources and memory, rendering it unsuitable for a low-latency routing system.

To bypass this, we implement a Sparse Variational Gaussian Process. Rather than utilizing the entire training dataset to compute a covariance matrix, a small, fixed set of summary variables known as *inducing points*  $Z$  are introduced by the SVGP. For our system, we initialize  $M = 100$  inducing points, sampled directly from the compressed training data  $X_{\text{train}}$ .

We are able to reduce the computational complexity of the Gaussian Process from  $O(N^3)$  to  $O(NM^2)$  by conditioning the Gaussian Process on these  $M$  inducing points, rather than the full  $N$  dataset.

We configured these inducing points to be dynamic. During the training phase, the coordinates of the inducing points are learnable parameters. During the iterations of the optimizer, these points are dynamically repositioned within the 10-dimensional latent space to effectively summarize the boundaries between the successful and failed SLM executions.

In order to maintain the mathematical stability of the system during this optimization process, the SVGP utilizes a ***Cholesky Variational Distribution***. Due to the covariance matrices within Gaussian Processes representing uncertainty, their values must mathematically remain positive semi-definite; negative variance is not defined. However, hardware-accelerated training can occasionally introduce microscopic rounding errors that push these values below zero, causing model failure. The Cholesky distribution acts as a mathematical safety net. It represents the covariance matrix  $\Sigma$  as the product of a root matrix  $L$  and its transpose:

$$\Sigma = LL^T$$

Due to the optimization now taking place on the root matrix  $L$  instead of the raw covariance matrix, the system inherently squares the internal values, effectively guaranteeing the resulting matrix remains positive semi-definite and preventing optimization crashes.

### A.2 The Kernel and Automatic Relevance Determination

The SVGP utilizes a covariance module to evaluate the semantic similarity between the user's prompts and the inducing points. The RBF kernel acts as the foundational component for the module.

A standard RBF kernel assumes that every dimension in the feature space is equally important. While the PCA compression isolated the 10 dimensions with highest overall variance, not all of that variance is directly correlated with the competence boundaries of the Small Language Model. To resolve this, the RBF kernel was configured with ***Automatic Relevance Determination (ARD)***. ARD assigns independent, learnable length scale parameters to each of the 10 individual dimensions. During the gradient descent process, the optimizer scales the individual parameters based on their predictive utility. If a particular PCA axis contains information deemed vital to SLM

failure prediction, the optimizer shrinks its length scale, increasing the GP’s sensitivity to small changes along that axis. Conversely, if an axis contains mostly noise, the optimizer will stretch its length scale out, effectively ignoring the impact of that dimension.

Finally, to properly calibrate the magnitude of the GP’s uncertainty, this ARD-enabled RBF kernel is wrapped in a *Scale Kernel*. Due to the RBF kernel strictly outputting correlation values between 0 and 1, the Scale Kernel will introduce a global variance parameter. This allows a better representation of the true mathematical spread of the latent space, by more accurately scaling the covariance matrix.

### A.3 Likelihood and Optimization

Due to our routing architecture acting as a binary gatekeeper, the GP cannot output raw continuous values. Passing the raw outputs of the GP through a *Bernoulli Likelihood* module allows for the required binary output format. Specifically, this squashes the multivariate Gaussian outputs into a strict probability space between 0 and 1.

The entire architecture of the SVGP, including the inducing point locations, the global variance, and the ARD length scales, is optimized simultaneously. The Evidence Lower Bound (ELBO) is maximized by our configured Adam optimizer. By doing this, the optimizer inherently balances two competing objectives: the model is pushed to fit the training data accurately, while simultaneously applying penalties to prevent the model from overfitting or becoming overly confident in sparsely populated regions of the latent space.